A MAP THAT IS INJECTIVE AND UNRAMIFIED AT A PRIME IS ETALE AFTER A PRINCIPAL LOCALIZATION, OVER A GEOMETRICALLY UNIBRANCH DOMAIN

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 31762-31774 (canonical tag 0GSC) and the counted proof is the proof environment at source lines 31776-31815, which immediately follows it and is its own proof.

Editorial note. The proof refers to four results of other chapters, namely the locally standard form of unramified maps, etale algebras over a field, the element separating minimal primes and normal algebras for etale maps, and to three results of this chapter among which are the two branch-count statements and the lemma on ideals meeting the source domain. All are carried as navigation stubs with their roles recorded in the reasons.

Editorial note. The proof contains the slip that the minimal primes of the etale cover are lie over the generic point of the base, which is copied exactly as printed and is not corrected.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- An unramified ring map of finite type is locally standard etale, so after a principal localization there is an etale map onto the ring with a surjection from it. invoked at proof lines 31777-31779. Replaces the given map by a surjection from a standard etale algebra, which is the setting in which the injectivity argument is run.
- An etale algebra over a field is a finite product of fields, and an etale map is flat with the same minimal primes as the base, so the number of minimal primes is finite. used at proof lines 31781-31788. Identifies the fibre over the generic point and shows that the etale cover has finitely many minimal primes.

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- Over a geometrically unibranch local ring the localization of an etale algebra at a prime is a domain, so the prime contains exactly one minimal prime. used at proof lines 31788-31797. Justifies the choice of an element that vanishes on all but one minimal prime so that the cover becomes a domain after localization.

Source reference. Original proposition 0395: sources/stacks-chow/tags.txt, source lines 4260–4260. The reference produces the standard etale cover with a surjection onto the given ring, used at proof lines 31777-31779; the reduction and the injectivity argument that follow are inside the counted span.

Source reference. Original lemma 00U3: sources/stacks-chow/tags.txt, source lines 1097–1097. The reference identifies the fibre of an etale algebra over a field as a finite product of fields, used at proof lines 31784-31786 and again at the end for the residue fields; the reduction built on it is inside the counted span.

Source reference. Original lemma 0DQ1: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31596–31596. The reference is one of the two applied at proof lines 31790-31791 to see that the localization of the etale cover at the prime is a domain; the use of that domain property is inside the counted span.

Source reference. Original lemma 0C37: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31566–31566. The reference is the second of the two applied at proof lines 31790-31791 for the same purpose; it is a pointer to a same-chapter result and everything drawn from it is inside the counted span.

Source reference. Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 508–508. The reference supplies the element that lies in all minimal primes except one, used at proof lines 31794-31795; the localization that consumes it is inside the counted span.

Source reference. Original lemma 0GS5: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31702–31702. The reference is applied at proof lines 31807-31809 to the domain map of localizations to find a nonzero element of the base mapping to zero, which is the contradiction; the setup of that contradiction is inside the counted span. The lemma itself is the subject of a separate card of this lane with disjoint coordinates.

Source reference. Original lemma 07QK: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10533–10533. The reference is cited at proof lines 31783-31784 to see that the etale cover of the domain is reduced, which the reduction to a domain then uses; the reduction itself is inside the counted span. The author cites it with the label as printed, which lacks the chapter prefix of the other Algebra references in the same proof.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $\mathfrak{q} \subset B$ be a prime ideal lying over the prime $\mathfrak{p} \subset A$. Assume*

(1) *A is a domain,*
(2) *$A_\mathfrak{p}$ is geometrically unibranch,*
(3) *$A \to B$ is unramified at $\mathfrak{q}$, and*
(4) *$A_\mathfrak{p} \to B_\mathfrak{q}$ is injective.*

Then there exists a $g \in B$, $g \notin \mathfrak{q}$ such that B_g is étale over A.

Proof. By Algebra, Proposition 0395 after replacing B by a principal localization, we can find a standard étale ring map $A \to B'$ and a surjection $B' \to B$. Denote $\mathfrak{q}' \subset B'$ the inverse image of $\mathfrak{q}$. We will show that $B' \to B$ is injective after possibly replacing B' by a principal localization.

In this paragraph we reduce to the case that B' is a domain. Since A is a domain, the ring B' is reduced, see Algebra, Lemma 07QK. Let K be the fraction field of A. Then $B' \otimes_A K$ is étale over a field, hence is a finite product of fields, see Algebra, Lemma 00U3. Since $A \to B'$ is étale (hence flat) the minimal primes of B' are lie over $(0) \subset A$ (by going down for flat ring maps). We conclude that B' has finitely many minimal primes, say $\mathfrak{r}_1, \ldots, \mathfrak{r}_r \subset B'$. Since $A_\mathfrak{p}$ is geometrically unibranch and $A \to B'$ étale, the ring $B'_{\mathfrak{q}'}$ is a domain, see Lemmas 0DQ1 and 0C37. Hence $\mathfrak{q}' \supset \mathfrak{r}_i$ for exactly one $i = i_0$. Choose $g' \in B'$, $g' \notin \mathfrak{r}_{i_0}$ but $g' \in \mathfrak{r}_i$ for $i \neq i_0$, see Algebra, Lemma 00DS. After replacing B' and B by $B'_{g'}$ and $B_{g'}$ we obtain that B' is a domain.

Assume B' is a domain, in particular $B' \subset B'_{\mathfrak{q}'}$. If $B' \to B$ is not injective, then $J = \mathrm{Ker}(B'_{\mathfrak{q}'} \to B_\mathfrak{q})$ is nonzero. By Lemma 0GS5 applied to $A_\mathfrak{p} \to B'_{\mathfrak{q}'}$ we find a nonzero element $a \in A_\mathfrak{p}$ mapping to zero in $B_\mathfrak{q}$ contradicting assumption (4). This finishes the proof. $\square$